# 1. Clinical Question & Core Hypothesis

## Clinical Question

The Ontologist highlighted three complementary biomarker axes in NMIBC—local immune organization from BCG, urine DNA methylation diagnostics, and circulating ADNP mRNA/CTCs—raising the question: why do some tumors with apparent local immune engagement (e.g., TLS/TILs) still fail to respond to intravesical BCG? 2) Tier 1 paths used: I built on (a) the BCG → immune-readout/trial axis (mechanistic therapy stimulus → local immune organization) and (b) the ADNP liquid■biopsy path (ADNP mRNA in CTCs as a tumor-intrinsic circulating signal) because these directly offer a mechanistic interaction between tumor-intrinsic programs and local immune architecture. 3) Tier 2 evidence that supports the claim: Immunoscore-like metrics and TP53-driven subtype biology provide independent immune-context and tumor■intrinsic proliferative/genomic■instability modifiers that plausibly alter TLS function; systemic/metabolic markers (HDL-C, IGF1) and cross■cancer methylation signals (SDC2/TFPI2) offer additional contextual modifiers. 4) Endpoint choice rationale: I selected therapy_response (response to intravesical BCG) because the core translational question from the Ontologist concerns why BCG succeeds or fails and because trials/clinical decision-making around BCG are the most actionable application of these biomarker axes. 5) Deliberate exclusions: I excluded urine■DNA methylation panels from the core hypothesis to keep the mechanistic focus tight (immune architecture vs tumor ADNP programs) and relegated systemic/metabolic and cross■cancer methylation signals to secondary_modifiers to be tested as effect modifiers rather than central drivers?

## Working Hypothesis

In NMIBC, ADNP-high tumors exhibit impaired local antigen presentation and TLS maturation, despite the presence of TLS-like aggregates, leading to suboptimal intravesical BCG–driven T cell priming. Clinically, ADNP-high status will attenuate therapy_response to BCG in TLS-rich tumors, whereas combining TLS quantification with ADNP expression will improve prediction and guide alternative or adjuvant therapies.

| Element | Value |
|---|---|
| Imaging feature | TLS-like lymphoid aggregates |
| Anchor gene(s) | ADNP |
| Primary endpoint | therapy_response |

# 2. Why This Gene — Not Others

## Primary hypothesis gene(s)

**ADNP:** ADNP liquid■biopsy path (ADNP mRNA in CTCs as a tumor-intrinsic circulating signal) because these directly offer a mechanistic interaction between tumor-intrinsic programs and local immune architecture.

## Alternative / secondary genes considered

| Tier | Gene | Role / Reason excluded from core |
|---|---|---|
| Secondary | TP53 | Supporting modifier from knowledge graph |

| | | |
|---|---|---|
| Secondary | Immunoscore-like immune metrics | Supporting modifier from knowledge graph |
| Secondary | SDC2 | Supporting modifier from knowledge graph |
| Secondary | TFPI2 | Supporting modifier from knowledge graph |
| Secondary | HDL-C | Supporting modifier from knowledge graph |
| Secondary | IGF1 | Supporting modifier from knowledge graph |

## 3. Clinically Testable Predictions

**Prediction 1:** and methylation markers (SDC2/TFPI

**Prediction 2:** could influence overall immune readiness and tumor behavior as effect modifiers. Experimental approach:

**Prediction 3:** Collect NMIBC tissue (pre-BCG) and measure ADNP expression at mRNA and protein levels in tumor cells and CTCs; quantify TLS-like aggregates by multiplex immunohistochemistry (markers for T cells, B cells, helper T cells, and follicular dendritic cell networks) and assign an Immunoscore-like metric.

## 4. Minimum Viable Validation Plan

### A. Retrospective Cohort Analysis

**Data source:** TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

### B. Tissue-Level Validation

**Methods:** Multiplex IF / IHC / spatial transcriptomics

**Key readouts:** - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

**Goal:** Confirm that the proposed mechanism operates at the spatial / tissue level.

### C. Functional Validation

**Models:** Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

**Perturbations:** Genetic (KO/rescue) or pharmacologic modulation of anchor gene

**Readouts:** T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

**Goal:** Establish causal relationship between anchor gene activity and immune efficacy.

# 5. Strengths, Caveats & Alternative Directions

## Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

## Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

## Alternative Directions

- TP53: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- Immunoscore-like immune metrics: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.